

Methods for Attenuation of Unmanned Aerial Vehicle Noise

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Abstract - Unmanned aerial vehicles (UAVs) like drones (quadcopters, hexacopters, octocopters etc.) can be a source of a significant acoustic noise. High noise makes them less suitable for use in densely populated urban areas, particularly during take-off, landing and low level flight, due to the noise annoyance. This paper reviews methods and proposes some concepts for attenuation of UAV noise. Both passive and active solutions are considered. Solutions with piston engine silencer, Q-tip propeller, more propeller blades, absorptive and reflective barrier, ducted propeller, sound absorbing ducts, synchrophaser, multichannel active noise cancellation (ANC) system with secondary sources circularly arranged around the propeller and the system with magnetically excited propeller blades are mentioned.

Keywords - *unmanned aerial vehicle, noise, propeller, sound absorption, active noise control*

I. INTRODUCTION

An Unmanned Aerial Vehicle (UAV) is an aircraft with no pilot on board. It is also referred as an unmanned aircraft system (UAS) or more frequently, drone. There are many civilian, commercial, military, and aerospace applications for UAVs, [1,2]. UAVs often produce a significant amount of acoustic noise that interferes with their operation in urban and other inhabited areas, particularly during take-off, landing and low level flight, [3-6]. Annoyance may not be necessary caused just by the UAV noise level, but by the specific noise spectrum. Methods for reduction of UAV noise can be broadly categorized as passive and active. Passive methods are well known and proven, don't use additional energy, but are sometimes bulky and not suitable for counteracting low frequency noise, [7]. On the other hand, active methods promise much higher noise attenuation at lower frequencies, [8-10]. However, such concepts are mostly novel, unproven in the context of UAVs and require an additional energy source. The combination of these two approaches may be a way to go.

II. TYPES OF UAVS

UAVs may be broadly categorized as a fixed wing and rotary wing devices. First UAVs were a fixed wing, they were extension of remotely controlled small aircrafts, equipped with additional electronics for navigation and reconnaissance, [1,2]. One example of a heavy fixed wing UAV is illustrated in Fig. 1, (General Atomics MQ-9). Fixed wing UAVs are powered by a small piston (gasoline or diesel) or even turboprop engine, often in a pusher configuration, and rarely by an electric motor (due to the



Figure 1 Fixed wing UAV (MQ-9) with turboprop engine (GA-ASI)

limited autonomy), [11-13]. Noise reduction for such vehicles is mostly directed toward silencing the engine and propeller. In case of a fixed wing UAV, the propeller can be front or rear mounted. In case of the rear mounted propeller and turboprop engine, there is an additional problem of exhaust going through the blades, producing nasty noise. On the other spectrum of UAVs are rotary wing devices, either a small helicopters or multirotor devices, Figs. 2-4. Multirotors can fly in every direction,



Figure 2 Quadcopter (DJI)



Figure 3 Hexacopter (Yuneec)



Figure 4 Octocopter (DragonRC)

horizontally and vertically, as well as hover in a fixed position. Common configurations are quadcopters (with four rotors), Fig. 2, hexacopters (with six rotors), Fig. 3 and octacopters (with eight rotors), Fig. 4. Quadcopters and hexacopters are more simple and common in use, with quadcopter configuration, particularly becoming popular for small UAVs. They are of much simpler design, but less redundant in case of loss of motor. Octacopters, however, are far more powerful machines than quadcopters or hexacopters.

According to the used propulsion type, multirotor UAVs can be electric (using batteries to power their flight) or can use piston engine that very rarely drive propellers, but can drive a generator for electric motors, [14].

Regarding the weight they can be categorized as light drones (< 25 kg, < 150 kg) and heavy drones (> 150 kg).

III. SOURCES OF NOISE

The sources of noise for a typical drone are shown in Fig. 5 (some sources not applicable for a particular type of drone), [15]. A dominant source of noise is a propulsion system, i.e., combination of engine or motor and propeller.

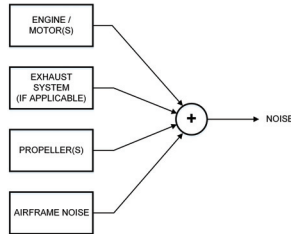


Figure 5 Contributions to aircraft noise

TABLE I. NOISE LEVELS FOR SMALL DRONES AT CLOSE DISTANCE

Drone	Noise level (dB) ¹⁾
Mavic Pro	98
Mavic Pro Platinum	93
Mavic Air	99

¹⁾ measured at 15 cm distance below the hovering drone

Even small drones may produce significant noise at close distance, see Table I. Mavic Pro Platinum produces 93 dB (due to special tip vortex design of the Platinum) and Mavic Air 99 dB (has a higher RPM), [6] Larger drones produce even higher noise levels. In general the lighter the drone, the quieter it will be. It also depends on the prop size in relation to the weight, motor RPM, the prop design and the amount of prop/body overlap. An example of the noise spectrum for two popular small drones is given in Fig. 6 (distance of several feet). It is presence of high frequencies that add to subjective annoyance ratings.

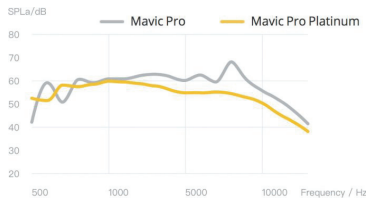


Figure 6 Noise spectrums of two popular drones (DJI)

Not all sources of noise are equally important. Some sources by far dominate the noise level, while others just symbolically contribute to the overall noise. Anything that is several dB lower than the other sources can be

discarded, as it contributes negligibly to the overall noise level. Levels of sound pressure decrease with the distance from the sound source with 6 dB per distance doubling (depending on atmospheric and geographic conditions, can be a 3 to 7 dBA range), (1), where L_1 and L_2 are sound pressure levels in dB at distances d_1 and d_2 from the sound source, [7,16].

$$L_2 - L_1 = 20 \log \frac{d_2}{d_1} \quad (1)$$

A. Piston Engine Noise

Piston engine noise is produced by pressure pulses at the intake and exhaust during the engine four cycles. The engine exhaust noise originates at the exhaust tailpipe opening. Noise from a small piston engine may be far in excess of 85 dB. Cylinder firing rate, f_{CFR} , is dependent on the rotational speed N_{RPM} , (RPM, rotations per minute) that is for a two stroke engine given by (2), [15]:

$$f_{CFR} = \frac{N_{RPM}}{60} \quad (2)$$

and for a four stroke engine is given by (3):

$$f_{CFR} = \frac{N_{RPM}}{120} \quad (3)$$

The engine firing rate, f_{EFR} , is dependent on the f_{CFR} and the number of cylinders N_C , (4):

$$f_{EFR} = N_C f_{CFR} \quad (4)$$

The perceived main frequency of the noise radiating from a 2 stroke engine at 10,000 RPM is 166 Hz, while from a 4 stroke engine at the same RPM is just 83 Hz.

B. Turboprop Engine Noise

Turboprop engines are turbine engines driving a propeller. The turbine engine noise is caused by the whining noise that comes from the intake, and the low frequency noise that comes from the turbine exhaust. Exhaust noise consists from both combustion noise, and the noise produced by the differential between the high velocity jet gases and the low speed outside air. The pressure/temperature difference of the air at the exhaust is the major contributor to a turbine engine noise.

C. Electric Motor Noise

Noise and vibration of an electric motor are caused by the structural deformation due to electromagnetic forces, moving parts (imbalance, gears, and bearings) and airflow through the motor, [17-20]. Today, mostly used are brushless DC (BLDC) motors, known for their wide application and energy efficiency. Stator excitation in BLDC motor drives can excite mechanical resonances of the motor and produce an acoustic noise similar to switched reluctance drives. Noise levels and spectrum are shown in [17]. Most popular BLDC motors use three phases. The motor speed



Figure 7 Set of electrical motors for a drone (HobbyMate)

control is accomplished using the PWM technique. Basically two methods are available for controlling the speed of a BLDC motor: sensor and sensorless control. Electronically commutated BLDC motor use information obtained from Hall sensors. Motors have a KV rating that relates to how fast it will rotate for a given voltage. Electric motors for powering drones are operating at quite high RPMs, [21]. RPM range of motors in Fig. 7 begins at 10,350 RPM and goes up to 41,400 RPM (unloaded).

D. Propeller Noise

Propeller noise is composed of tonal and broadband components. Sources of propeller aerodynamic noise are shown in Fig. 8. The greatest noise, in terms of directivity, is in the propeller plane. The basic frequency f_1 or f_{BPF} (Blade Pass Frequency), is the product of the propeller rotational speed and the number of propeller blades (5), [15]:

$$f_1 = f_{BPF} = \frac{N_{RPM} \cdot N_B}{60} \quad (5)$$

where is:

f_1, f_{BPF} basic frequency of tonal propeller component
 N_{RPM} propeller rotational speed (rotations per minute)
 N_B number of propeller blades.

The tonal components contain basic frequency f_1 and its harmonics f_n with spectral components shown in Fig. 9, (6), [15]:

$$f_n = N f_1 \quad N = 1, 2, \dots \quad (6)$$

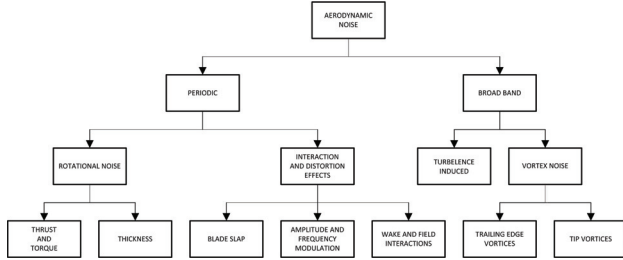


Figure 8 Sources of propeller aerodynamic noise, according to [23]

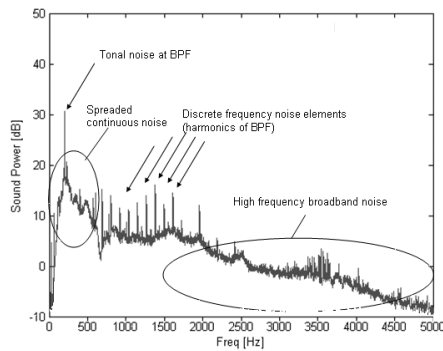


Figure 9 Noise spectrum of a propeller, modified from [24]

The propeller tip speed is determined by (7):

$$v = \frac{DN_{RPM}\pi}{60} \quad (7)$$

where is

v propeller tip speed
 D propeller diameter
 N_{RPM} propeller rotational speed

Propeller tip speeds (ms^{-1}) for various small propeller diameters and rotational speeds are shown in Table II.

TABLE II. PROPELLER TIP SPEEDS FOR VARIOUS PROPELLER DIAMETERS AND ROTATIONAL SPEEDS ¹⁾

Propeller diameter (inch)	Rotational speed (RPM)								
	2000	3000	4000	5000	6000	7000	8000	9000	10000
3	7.97	11.97	15.96	19.95	23.93	27.93	31.92	35.91	39.90
4	10.64	15.96	21.28	26.60	31.92	37.24	42.56	47.88	53.20
5	13.30	19.95	26.60	33.25	39.90	46.55	53.20	59.85	66.50
6	15.96	23.94	31.92	39.90	47.88	55.86	63.84	71.82	79.80
7	18.62	27.93	37.24	46.55	55.86	65.17	74.48	83.79	93.10
8	21.27	31.92	42.56	53.20	63.84	74.48	85.12	95.76	106.4
9	23.94	35.91	47.88	59.85	71.82	83.79	95.76	107.7	119.7
10	26.60	39.90	53.20	66.50	79.80	93.10	106.4	119.7	133.0
11	29.26	43.89	58.52	73.15	87.78	102.4	117.0	131.7	146.3
12	31.91	47.88	63.84	79.80	95.76	111.7	127.7	143.6	159.6

¹⁾ yellow color denotes noisy speeds $>100 \text{ ms}^{-1}$ ($\approx 0.3 \text{ M}$)

Small drones use exclusively fixed pitch propellers, and changing its thrust is accomplished by changing the RPM of the electric motor that are driving them.

E. Airframe Noise

It is mostly negligible at low speeds and in comparison to other sources of noise (like a propeller and engine).

F. Noise Levels From Multiple Sources

Summation of sound pressure levels from multiple incoherent sound sources (e.g., propellers with different RPMs), [7], is defined by (8), n is the number of sources and L_n is the sound pressure level of the source n in dB:

$$L_x = 10 \log_{10} \left(10^{\frac{L_1}{10}} + 10^{\frac{L_2}{10}} + \dots + 10^{\frac{L_n}{10}} \right) \quad (8)$$

The increase of noise level ΔL from multiple equally loud incoherent sound sources is defined by (9).

$$\Delta L = 10 \log_{10}(n) \quad (9)$$

Important are the contributions from the most prominent noise sources, other noise sources (if not numerous) can be neglected. The rise of the noise level by the contribution of more equally loud incoherent sound sources is illustrated in Table III.

TABLE III. CONTRIBUTION OF MORE EQUALLY LOUD INCOHERENT SOUND SOURCES

Sources	Δ noise level (dB)
1	0
4	6,0
6	7,8
8	9,0

IV. PASSIVE METHODS FOR NOISE REDUCTION

Following passive methods may be used for reducing the noise of UAV:

- Piston engine silencer (muffler),
- Modification of blade airfoil shapes (e.g., Q-tip propeller, saw tooth serrated trailing edge)
- Propeller with more blades (three, four, five, six ...)
- Absorptive and reflective barriers (sound reflectors)
- Ducted (shielded) propeller with sound absorbers (absorbing surface materials and metasurface absorber based on a Helmholtz resonator)

A. Piston Engine Silencer (Muffler)

Piston engine may be silenced by use of absorptive (dissipative) and/or reactive silencer (muffler), [25,26], as in Fig. 10 and Fig. 11. Achieved noise levels for popular engine/muffler combinations are given in [27]. An



Figure 10 Small two stroke piston engine with a silencer (muffler)

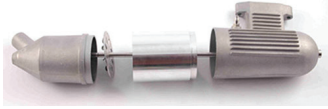


Figure 11 Interior of small reactive silencer

additional step for larger engines would be use of a silencer at the engine intake. Also, care should be taken that engine doesn't excite fuselage resonances. Results of application of silencer on a small aircraft are given in [28].

B. Modifying Blade Shape (Q-Tip Propeller)

Blade shape and airfoil section shape influence to a small degree (a few dB) the noise level produced by a propeller. Particularly popular is a Q-tip propeller, first developed in the 1970s, that uses bent tips (90°) to reduce the tip noise, [29]. It reduces the airflow, in the same way as winglets reduce wing tip vortices. Bent tip propeller inspired by the same concept is shown in Fig. 12. The influence of propeller blade geometry to propeller noise level is investigated in [29,30]. The effect of saw tooth serrated trailing edge propeller is analyzed in [31].



Figure 12 Q-tip style propeller (DJI Mavic Pro)

C. Use of Propellers with More Blades

Dual blade propellers are the most commonly used in drones. Same thrust can be achieved with a propeller with the smaller diameter, but more blades (three, four, five, or even six). Example of six blade propeller for the drone is shown in Fig. 13. Increasing the number of blades leads to a lower tip speed, but also to somewhat decreased efficiency. The lower tip speed reduces the noise generated by a propeller. Propellers with more blades may require more effort to balance, but the results may be dramatic in terms of achieving very low vibration levels and much lower noise levels. Changes in noise levels with increasing the number of blades, achieved in one general aviation aircraft, according to [28] are shown in Table IV.



Figure 13 Three blade propellers for a drone

TABLE IV. NOISE REDUCTION CHANGE WITH INCREASING NUMBER OF BLADES

Blades	Noise reduction
two to three	-4 dB
three to four	-4.5 dB
four to five	-4.5 dB

D. Use of Absorptive and Reflective Barriers

Noise from the drones emanates in many directions, some causing more annoyance than others. Using the sound reflectors, the noise may be partially reflected upwards. Reflective materials have hard surfaces that reflect the sound, [7]. Materials with various degrees of combinations of absorption and reflections may also be used. Depending on the material density and how porous it is, some degree of absorption may also take place, converting unwanted noise to heat. Sound reflector could be placed at an angle below the propellers, at the outer edge of the propeller ring.

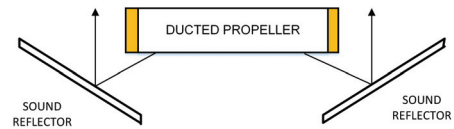


Figure 14 Use of sound reflector for reflecting propeller noise

With the use of sound reflectors, Fig. 14, propeller noise is partially reflected up, where it generally has no environmental impact.

E. Ducted (Shielded) Propeller

Ducted (shielded) propeller is an arrangement where the propeller, is mounted within a cylindrical shroud or duct propeller, Fig. 15, [32]. It is more efficient in producing a thrust than a conventional propeller of similar diameter. Often they also provide (but not always, due to interaction with the casing) a lower noise (shield the blade noise, and reduce the tip speed), [33,34]. Duct also offers great protection from propeller blades (safety factor), Fig. 16, [35]. There is an opportunity for acoustic treatment of the surfaces of the duct with sound absorptive material. Thickness of the material influences the absorption, but thicker material requires a duct with a slightly greater diameter ($\frac{1}{4}$, $\frac{1}{2}$ inch or more). Propeller duct also offers the possibility for the use of a circularly shaped perforated absorber (acoustic liner) and metasurface absorber based on Helmholtz resonator, Fig. 17, that may be quite capable (several dB) of reducing low and mid frequency noise, [36].



Figure 15 Ducted (shielded) propeller (Aerotestra MK10)



Figure 16 Futuristic IFO passenger drone concept with ducted (shielded) propellers (Jet Capsule), [35]

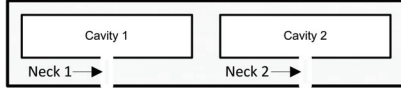


Figure 17 Metasurface absorber

V. ACTIVE METHODS FOR NOISE REDUCTION

Active methods for noise reduction are, due to the complexity, intended (but not restricted) for larger drones that produce high noise levels. Such methods include:

- Synchrophaser for reducing the noise and propeller beats (optional)
- Active noise control for reducing the propeller noise radiated into the environment using speakers in a circular arrangement placed around the propeller
- Propeller blades with small magnets that vibrate in the electromagnetic field produced by the coil in the propeller frame

A. Synchrophaser

A synchrophaser is a device that precisely matches the RPM and the phase of all the propellers, Fig. 18, [37]. It compares the propeller to propeller relative positions of the propeller blades within their individual arcs and adjusts their angle to their optimum relative positions, [38,39]. The noise from one engine-propeller can be tuned to provide a canceling acoustic wave to reduce the noise generated from another engine-propeller operating on the same aircraft. As such, synchrophaser can be used for noise cancellation on multi engine/motor drones with variable pitch propellers rotating at the same RPM. The Air Force Research Laboratory (AFRL) has developed a second generation phase-locked-loop (PLL) digital synchrophaser in 1994. The synchrophaser achieved 3-5 dB reduction in exterior noise levels on a four engine C-130 aircraft, [40].

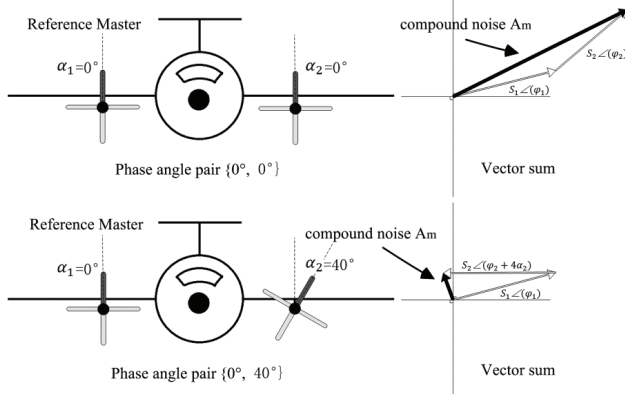


Figure 18 Synchrophaser operation and noise attenuation, [35]

B. Active Noise Control

Active noise control (ANC) is a method for the reduction of unwanted noise by generating an acoustic signal that interferes with the offending noise (use of destructive interference), [8,9]. Use of ANC has achieved a significant reduction of tonal noise components.

• Multichannel ANC

The multichannel ANC, with 12 speakers (number may vary), proposed by the author of this paper, can be used for active noise reduction, Fig. 19. Speakers are located in a circular arrangement around the propeller on the outside surface of the propeller duct, radiating sound at an angle (e.g. 45°) and driven by the multichannel ANC controller.

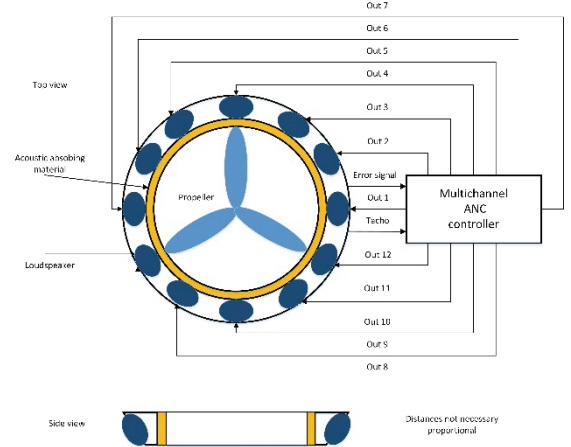


Figure 19 Multichannel ANC of propeller noise

Each of acoustic channels is driven by the waveform synthesis ANC, Fig. 20, supplying the same waveforms, but at different phase angles (e.g., in case of 12 channels, each channel is shifted by 30° from the previous one). In case of three blade propeller 12 channels can be grouped into 3 groups of 4 channels (channels in the same group are same). The waveform synthesizer, [41,15], has stored waveform samples (10), L is the period length in samples:

$$w_l(n), l=0,1, \dots, L-1 \quad (10)$$

Canceling noise waveform for the secondary speaker at the moment n is given by (11) and (12):

$$y(n) = w_{j(n)}(n) \quad (11)$$

$$j(n) = n \bmod L \quad (12)$$

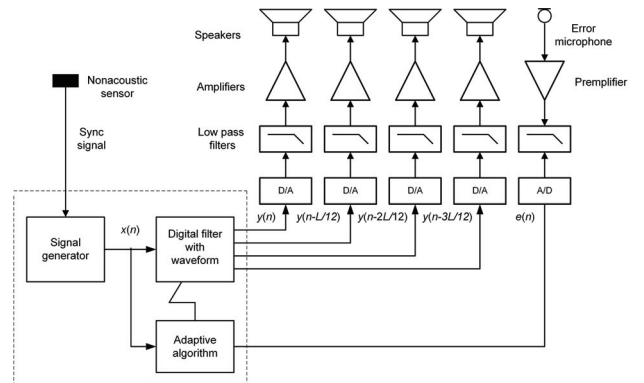


Figure 20 Multichannel ANC with four channels (delayed waveforms)

C. Magnetically Excited Propeller Blades

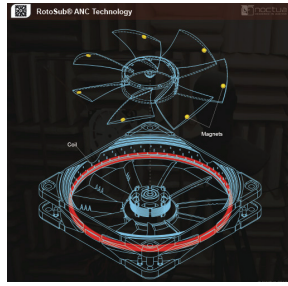


Figure 21 Blades vibrating in electromagnetic field (Noctua RotoSub®)

Very interesting approach for silencing computer cooler noise was proposed by Noctua and patented as RotoSub® technology, Fig. 21, [42]. Although primarily intended for computer cooling fans, a similar approach could probably be applied to small drone propellers as well. Switching on/off the electricity to the coil placed within the propeller frame interacts with small magnets inside the blade tips providing minute modulations of the rotary movements of fan blades. This allows the fan itself to emit the anti-noise signal through the blades, practically at the same position as the originating noise making it easier to precisely match its phase and rotation pattern.

VI. CONCLUSION

Drones have numerous applications that require them to operate in urban areas where noise annoyances are possible. To reduce occurrences of such events it may be necessary to implement some strategies for noise reduction, particularly during low level operations. Passive methods have its attraction in simplicity, reliability, low cost and require no additional power. Reduction of low frequencies may require ANC methods that can be, due to the power requirements, switched on only during take-off, landing and low level flight. Combining passive and active noise reduction methods to achieve greater noise reduction may be a feasible approach. Each of listed methods would contribute little to the overall noise reduction, but combined reduction may match the desired noise reduction target.

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